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Applied Stats I
Exam One

Instructions: Answer the following problems in the space provided. Please be sure to show your work to receive full credit. Do your best to reduce your calculations as much as possible. Below, you are provided with z and t score tables.

Table 1: Z-score table from cumulative standardized normal distribution.

	$\alpha = .01$	$\alpha = .02$	$\alpha = .03$	$\alpha = .04$	$\alpha = .05$
z-score	1.9649	1.9656	1.9664	1.9671	1.9678

Table 2: Critical Values of t from t distribution.

Degrees of freedom	*Two-tailed*	10%	5%	2%
	One-tailed	5%	2.5%	1%
46		1.679	2.013	2.410
48		1.677	2.011	2.407
50		1.676	2.009	2.403

1 Descriptive statistics

The following table contains voter turnout rates from five counties in Ireland for the 2024 general election. The numbers indicate the percent of registered voters in each county that cast a ballot. Compute the mean, median, and standard deviation of these data.

Dublin Central	Kerry	Mayo	Louth	Waterford
67.4	59.2	72.6	51.3	66.0

$\bar{y} = 63.3$

med = 66.0

$sd = \sqrt{68.065} \approx 8.2$

$sd = \sqrt{\frac{272.25}{4}}$

$sd = \sqrt{68.065}$

51.3, 59.2, 66.0, 67.4, 72.6

$sd = \sqrt{\frac{(16.81 + 16.81 + 86.49 + 7.15 + 144)}{5-1}}$

2 Hypothesis testing

According to a union agreement, the mean income for all senior-level assembly-line workers in a large company equals €625 per week. A representative of a women's group decides to analyze whether the mean income μ for female employees matches this norm. For a random sample of 225 female employees, $\bar{y} = €600$ and $s = €70$.

Conduct a hypothesis test of the group's theory that women at this company are paid less than the average amount of €625. Use an $\alpha = .01$ level of significance.

Since $Z_{obs} = -4.9 \leq 1.9649$ and $\bar{y} = 625 \notin [590.834, 609.166]$ we reject the null hypothesis. Therefore, there is sufficient evidence within $\alpha = 0.01$ that female employees get paid less than 625.

$df = 224$

$\bar{y} = 600 \quad n = 225 \quad s = 70 \quad \mu_0 = 625 \quad \alpha = 0.01$

data is random, σ is unkown, but due to large sample size (n) we can use z-test distribution.

$$SE = \frac{70}{\sqrt{225}} = \frac{70}{15} \approx 4.66$$

Our hypotesis are: H_0 & $\mu \geq 625$, $H_A: \mu < 625$

$$Z_{obs} = \frac{600 - 625}{4.66} \qquad Z_{obs} = -\frac{25}{4.66} \approx -4.9$$

$Z_{\alpha=0.01} = 1.9649$

$-4.9 \leq 1.9649$ reject null

$CI = [590.834, 609.166]$

$CI = 600 \pm (1.9649)(4.66)$

$CI = 600 \pm 9.166$

$625 \notin [590.834, 609.166]$ reject null

3 Distributions

Answer the questions below. It is possible to answer with only two sentences and receive full credit (be clear and specific though).

a) Explain how a sample distribution, a sampling distribution, and a population distribution are different and how they are related.

A sample distribution is a subset of our population. When we talk about sample dist. we use estimators: $\bar{y}, s$.
A population distribution is when we know the distribution of our population, the population is our entire object of study. We use N and σ parameters.

A sampling distribution is the process where we draw samples distributi from our population. (For example for our CLT).

b) Your friend Anne told you that if you have a 95% confidence interval, you can be 95% sure that the true parameter of interest is contained in the interval. Explain why this is wrong, what a confidence interval is, and how we should interpret it.

Because a 95% CI doesn't mean that the true parameter has a 95% chance of being in the interval.

Instead it means that if we continue sampling samples from our population and repeat our test-statistic, 95% of the CIs will contain the true parameter.

A CI is a range of values where (for example here) 95% of the times we draw a sample, the true parameter will be inside this range of values.

If we set a 95% CI we say that the true parameter will be in this range of values 95% of the times.

4 Margin of error

This week, *Gallup* released a poll of 1,444 randomly selected adults. The poll asked people to indicate their approval level of President Trump. Respondents were allowed to give one of four responses: 1=Strongly approve, 2=Approve, 3=Disapprove, 4=Strongly disapprove. We are going to treat this as quantitative data. The mean response was 3.2 and the standard deviation was 1.9. Calculate a 98% confidence interval for the national approval of President Biden. Identify the margin of error for this poll.

$CI = [3.1902, 3.25]$

margin of error $= 0.0598$

$\bar{y} = 3.2$

$SE = \frac{1.9}{\sqrt{1444}} \approx \frac{1.9}{38} \approx 0.05$

$sd = 1.9$

$n = 1,444$

$\alpha_2 = 0.04$ $\quad 2(1-.98) = 2(0.02) = 0.04$

$z_{\alpha=0.04} = 1.9671$

$3.2 \pm (1.9671)(0.05)$

3.2 ± 0.098

$3.2800 - 3.1902 = 0.0598$

$3.2 + 0.05 = 3.25$

3.1902

5 Definitions

Define the following 4 terms. You can earn full credit with just one or two sentences, but please be clear and thorough.

Central limit theorem

For large random, independent samples, $\bar{Y}$ is approximately normal even if Y is not.
No matter which distribution $\bar{Y}$ is, if we continue to draw samples from our population (sampling) we will see how $\bar{Y}$ will become normally distributed.

$$\bar{Y} \approx N\left(\mu \frac{\sigma^2}{n}\right)$$

p-value

The p-value is what we use to reject or fail to reject the null. We compare it against α.

The p-value is a statistic we get that our test-statistic is in the distribution. It is how far away is our statistic in the distribution.

Standard error

The standard error is a measurment of how separated are the observations in the distribution. It is given by the standard deviation and the squared root of the sample size. This SE dictates how spread are the tails of the distribution.



Type II error

A Type II error is when we fail to reject H_0, but we should had rejected it. It is a false-negative. In reality the null was false.

For example in a court, a jury declares innocent, but in reality the person was guilty.

Calculating the probability of a Type II error is hard because we need to calculate the power of the test-statistic we are doing.

6 Short answer

You are surveying students at your college on whether they are registered to vote. You decide to conduct your survey by visiting 5 randomly chosen political science classes. In each class, you ask each student to stand up (in front of the whole class) and tell you, first, whether or not they are registered to vote, second, whether or not they think voting is an important civic duty. In the space below, identify two kinds of measurement error that might affect this study (A few sentences are enough to get full credit for this problem, but be clear and specific!).

1) Standing up in front of the whole class is not ideal because it might affect the responses of the stodents, by hearing what their classmates say. It would be better if the student were interviewed separately and without a chance to have contact with each other.

2) What is civic duty? This might be a thing that some students might have different definitions from others. We would need to define ourselfs what civic duty is and tell the students the definition for our study. Or change subject of study to results or turnout in elections.